

Bored Intern Assignment:

1. Problem Statement & Objective

Our marketing team relies on influencer partnerships to drive campaign success. Currently, the process of finding influencers, validating their metrics, logging their contact details, and sending outreach messages is entirely manual. This manual process is slow, prone to human error, and lacks scalable tracking. We need an automated solution to streamline this critical business function.

Your Objective: Design and develop an automated pipeline that identifies potential influencer partners on Instagram based on specific metric thresholds, extracts their public data into a centralized tracking spreadsheet, and automatically dispatches personalized outreach messages upon human approval.

2. Environment & Tools

You have flexibility in your technical approach. You may use a code-first approach (e.g., Node.js, Python) or a low-code orchestration tool (e.g., n8n, Make) combined with necessary APIs. However, your architecture must be clearly documented, and you must adhere to platform constraints.

3. Core Requirements & Workflow

Phase 1: Data Acquisition

The system must accept search parameters (e.g., target region, minimum/maximum follower count, specific niche keywords) and retrieve relevant Instagram profiles. For each targeted profile, the system must extract the following data points without requiring manual data entry:

- Username
- Full Name
- Follower Count
- Post Count
- Public Email Address (if available in bio or via action buttons)
- Public Phone Number (if available via business contact buttons or bio)
- Calculated Engagement Rate (%)
- Profile URL
- External URL (Link in bio)

Phase 2: Data Storage & Human-in-the-Loop Review

The extracted data must be automatically formatted and appended into a structured database or

spreadsheet (e.g., Google Sheets). This sheet will act as our review dashboard. There must be a specific column designated for approval (e.g., a dropdown with "Yes/No" or "Approved/Pending").

Phase 3: Automated Outreach Dispatch

The system must monitor the tracking sheet for changes in state. When a team member marks a specific influencer as "Yes" (Approved), the system should automatically trigger a personalized outreach message. This can be via an automated Email (preferred if an email is found) or a Direct Message.

4. Expected Deliverables

Deliverable	Description
Working Prototype	A functional demonstration of the end-to-end workflow processing at least 5 target profiles.
Source Code / Export	The raw code repository, script files, or exported workflow JSON (if using an orchestration tool).
Technical Readme	A brief document explaining your architectural choices, how you handled platform constraints, and instructions on how to run the system locally or in the cloud.

5. Evaluation Criteria

- **System Resilience:** How does your system handle missing data (e.g., a profile with no public email or phone number, or private accounts) without crashing?
- **Platform Constraints:** Social media platforms have strict security measures and rate limits. We will evaluate how you architect your solution to avoid blocks or timeouts.
- **State Management:** How does your system ensure that an approved influencer is only contacted exactly once, even if the workflow runs multiple times?
- **Code/Workflow Cleanliness:** Logical structuring of nodes, scripts, variables, and proper error catching.

6. Sample Data Schema (Output Expectation)

```
{  
  "username": "nihadsha_",  
  "fullName": "Nihad Sayyidali",  
  "followers": 92158,  
  "posts": 136,  
  "email": "Not specified",  
  "phoneNumber": "+91 92158 13618",  
  "engagementRate": "18.85%",  
  "profileUrl": "https://www.instagram.com/nihadsha_/", "externalUrl": "Not  
specified",  
  "approvalStatus": "Pending"  
}
```